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AgriSphere – Intelligent Digital Agriculture Ecosystem

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA0904 – Programming in Java

to the award of the degree of

**BACHELOR OF ENGINEERING
IN
INFORMATION TECHNOLOGY**

Submitted by

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Under the Supervision of

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "AgriSphere – Intelligent Digital Agriculture Ecosystem" has been carried out by A K Selva Prabhu (192421224) under the supervision of Dr. R. Thalapathi Rajasekaran, and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Information Technology programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I am, A K Selva Prabhu (192421224) of the Department of Information Technology, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "AgriSphere – Intelligent Digital Agriculture Ecosystem" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the planning, development, or documentation of this project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 04.09.2026

Signature of the Students with Names

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INDIVIDUAL CONTRIBUTION STATEMENT

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
A K Selva Prabhu (192421224)	User & Profile Management module	Designed and developed user registration, login, profile management and role-based access functionality.	Tested user authentication, profile operations and role-based access for different users.	Contributed to documentation of user management, system design and implementation details.	[20%]
Ganesh P (192421142)	Smart Digital Marketplace module	Designed and developed product listing, product browsing, search, filtering and marketplace functionality.	Tested product listing, searching, filtering, product selection and marketplace workflows.	Contributed to documentation of marketplace features, requirements and results.	[20%]
Dharshan A N (192525370)	AI Recommendation Engine module	Designed and developed AI-assisted crop, soil health and recommendation functionality using relevant agricultural parameters.	Tested recommendation and advisory inputs, generated outputs and verified functional behaviour.	Contributed to documentation of AI functionality, literature review and system analysis.	[20%]
Mohan Raj V D (192424151)	Order & Service Management module	Designed and developed cart, order, payment, service booking, tracking and notification workflows.	Tested order processing, service booking, payment-related workflow and status tracking.	Contributed to documentation of order/service management, testing and project results.	[20%]
Monesh S (192324027)	Analytics & Administration module	Designed and developed administrative monitoring, analytics and platform management functionality.	Tested dashboard information, administrative operations and analytics-related functions.	Contributed to documentation of administration, analytics, project management and overall report preparation.	[20%]
Total					100%

ABSTRACT

Agriculture supports the livelihood of a very large population, yet many of its stakeholders continue to depend on fragmented, non-digital processes. Farmers frequently struggle to find genuine buyers and obtain fair prices for their produce, buyers find it difficult to identify trustworthy farmers and quality products, and agricultural service providers such as equipment-rental or consultancy operators are not discoverable through any single platform. Existing digital agriculture solutions tend to address only one of these needs in isolation — sensor-based farm monitoring, crop recommendation, or online product listing — without integrating buying, selling, services, recommendations, and administration into one ecosystem. This creates a clear engineering problem: the absence of a centralized, role-based digital platform that connects every agricultural stakeholder under one system.

AgriSphere – Intelligent Digital Agriculture Ecosystem is proposed to address this problem. It is a centralized web-based platform, developed using Java-based programming and software-engineering principles, that connects Farmers, Buyers, Service Providers, and Administrators through a single system. The platform is organized into five major modules: User & Profile Management, which provides secure role-based registration and login; the Smart Digital Marketplace, through which farmers list produce and buyers search, compare and purchase it; an AI Recommendation Engine, which uses stated preferences and browsing behaviour to suggest relevant products; Order & Service Management, which handles cart, checkout, payment, service bookings and order tracking; and Analytics & Administration, which gives administrators dashboards to monitor users, products, orders and platform activity.

The methodology followed a systematic software-development approach: stakeholder requirements were identified and analyzed, a modular system architecture was designed around the five modules, the functionality was implemented and integrated, and the resulting system was verified against its functional requirements. The major result of the project is a working, integrated prototype in which farmers can manage digital profiles and products, buyers can discover and purchase produce, service providers can offer services, and administrators can monitor platform-wide activity — demonstrating that the identified modules can operate together as a single ecosystem rather than as isolated tools.

The principal conclusion is that a modular, role-based digital platform can meaningfully organize and simplify the interactions between the different stakeholders in the agricultural value chain, while remaining extensible toward future capabilities such as IoT-based farm monitoring, advanced predictive analytics, and mobile access.

Keywords: Digital Agriculture, Agricultural Marketplace, AI Recommendation System, Java Software Development, Role-Based Access Control, Agricultural Ecosystem Platform

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
API	Application Programming Interface	Software
JWT	JSON Web Token	Authentication
ORM	Object-Relational Mapping	Database
SQL	Structured Query Language	Database
SKU	Stock Keeping Unit	Inventory

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Agriculture is a sector that supports the livelihood of a very large population and plays a major role in economic development, yet many agricultural activities are still carried out using traditional, non-digital methods. Farmers often face challenges in finding genuine buyers and obtaining fair prices for their products, buyers find it difficult to identify trusted farmers and quality produce, and agricultural services such as equipment rental and consultancy are scattered across different, unconnected sources. These conditions create a need for a centralized digital platform capable of bringing agricultural buying, selling, services, and decision-support together in one place.

AgriSphere – Intelligent Digital Agriculture Ecosystem is developed as a response to this context. It is a Java-based, modular software platform that connects Farmers, Buyers, Service Providers, and Administrators through a single system, combining a digital marketplace, an AI-based recommendation engine, order and service management, and analytics and administration into one integrated ecosystem.

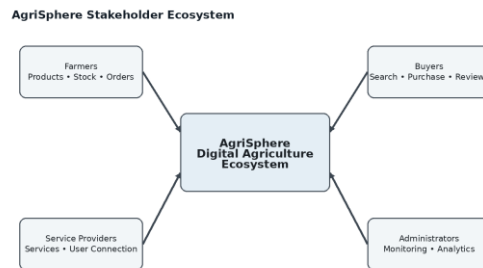


Figure 1.1: AgriSphere Stakeholder Ecosystem

1.2 Need for the Project

A centralized digital agriculture platform is needed because the stakeholders of the agricultural value chain currently interact through fragmented, largely offline channels. Farmers are affected because they lack a reliable, direct channel to reach buyers; buyers are affected because they cannot easily verify the trustworthiness or quality of produce from unfamiliar sellers; and service providers are affected because they have no common platform through which to advertise and deliver their services alongside the marketplace itself.

Existing limitations include the absence of a single system that combines product listing, buyer discovery, service booking, and recommendation-based decision support. Without such a

system, stakeholders must depend on manual searching, word-of-mouth, or multiple disconnected applications, which increases operational effort and reduces the overall efficiency of agricultural transactions. From an engineering standpoint, this establishes a clear significance for a role-based, modular platform that unifies these functions under consistent authentication, data-management, and workflow design.

1.3 Problem Statement

There is no single, centralized digital ecosystem that simultaneously connects Farmers, Buyers, Service Providers, and Administrators while integrating product listing and discovery, agricultural service booking, AI-assisted product recommendations, order and payment management, and platform-wide analytics. This problem involves several interacting engineering factors: it must support multiple distinct user roles with different permissions and workflows (role-based access), it must connect transactional functionality (marketplace, cart, payment, orders) with decision-support functionality (recommendations, analytics) within one consistent data model, and it must remain extensible enough to accommodate future capabilities such as IoT-based data collection without requiring a redesign of the core system.

1.4 Project Aim

The aim of the project is to design and develop AgriSphere, a centralized, modular, Java-based digital agriculture ecosystem that connects farmers, buyers, service providers, and administrators, and integrates a digital marketplace, an AI-based recommendation engine, order and service management, and analytics and administration within a single platform.

1.5 Project Objectives

- To design a role-based User & Profile Management module supporting secure registration, login, and access control for farmers, buyers, service providers, and administrators.
- To develop a Smart Digital Marketplace module through which farmers can list agricultural products and buyers can search, compare, and purchase them.
- To implement an AI Recommendation Engine that generates personalized product suggestions from user preferences and interaction history.
- To develop an Order & Service Management module covering cart operations, order placement, payment processing, service bookings, and order tracking.
- To implement an Analytics & Administration module that gives administrators dashboards and reports on users, products, orders, and platform activity.
- To test and verify that the integrated modules function correctly as a single connected platform.

1.6 Scope

Included: role-based registration and profile management for four user types; a digital marketplace for listing, searching, comparing, and purchasing agricultural products; an AI-based recommendation feature driven by user preferences and browsing behaviour; cart, order, payment, and service-booking workflows with order tracking and notifications; and administrative dashboards for monitoring users, products, orders, and sales.

Excluded: physical logistics and delivery-fleet management, IoT-based field sensing and hardware integration, and real-time agricultural commodity market-price feeds — these are identified as future extensions rather than part of the current implementation.

Assumptions and boundary conditions: the platform is assumed to be accessed through a web-based, internet-connected client; users are assumed to provide accurate profile and product information at the time of registration and listing; and the recommendation engine is assumed to operate on the preference and interaction data captured within the platform itself, without external market-data feeds.

1.7 Expected Engineering Outcome

The expected outcome is a working, integrated software prototype — the AgriSphere platform — in which the five modules operate together: a farmer can register, manage a profile, and list products; a buyer can discover, compare, and purchase those products or engage a service provider; and an administrator can monitor the resulting activity through a centralized dashboard. The prototype is intended to demonstrate the feasibility of the proposed modular architecture as a foundation for a scalable digital agriculture platform.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature was identified by reviewing recently published research and existing digital-agriculture products in the areas most relevant to AgriSphere's five modules: IoT-based farm monitoring, AI-based crop/yield recommendation, online agricultural marketplaces, and precision-agriculture platforms. Each source was examined for the specific agricultural function it addresses and the extent to which it integrates with other functions such as marketplace access or service booking.

2.2 Review of Existing Work

Research on smart agriculture monitoring using IoT and cloud computing concentrates on collecting and analyzing farm-sensor data but does not extend to an online marketplace connecting farmers and buyers. Research on AI-based crop recommendation and yield prediction focuses on generating agronomic recommendations but does not address product selling or business/order management. Online agricultural marketplace systems provide the reverse — product listing and selling — but typically lack AI-based recommendations and formal integration of agricultural service providers. Digital farming platforms centred on precision agriculture concentrate on farming technique optimization rather than connecting farmers, buyers, and service providers within one ecosystem.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Smart Agriculture Monitoring Using IoT and Cloud Computing (Patel & Singh, 2022)	Provides real-time farm/sensor monitoring	Focuses only on farm monitoring; no online marketplace connecting farmers and buyers	Confirms the need for a marketplace layer that AgriSphere adds on top of monitoring-style solutions
AI-Based Crop Recommendation and Yield Prediction System (Kumar et al., 2023)	Provides useful crop/yield recommendations	Does not support product selling or business management	Motivates AgriSphere's AI Recommendation Engine being combined with the marketplace, not offered in isolation
Online Agricultural Marketplace for Farmers (Sharma & Verma, 2021)	Supports online listing and selling of produce	Lacks AI recommendations and service-provider integration	Shows the gap AgriSphere closes by adding recommendations and a service-provider role
Digital Farming Platform with Precision Agriculture (Das & Roy, 2024)	Strong precision-farming technique focus	Does not connect buyers and agricultural services to the farming platform	Reinforces the need for an ecosystem that unifies farming, buying, and services rather than one function alone

2.4 Research / Engineering Gap

Across the reviewed approaches, a consistent gap emerges: existing solutions tend to address a single agricultural function — monitoring, recommendation, or marketplace listing — in isolation, rather than integrating farmers, buyers, service providers, AI-based recommendations, order and service management, and analytics into one connected digital ecosystem. No single reviewed system offers all of these functions together under a common, role-based platform.

2.5 Proposed Contribution

AgriSphere addresses this gap by combining a digital marketplace, an AI recommendation engine, order and service management, and analytics and administration within one platform, connected through a shared user and role-management layer. Rather than offering an isolated monitoring, recommendation, or listing tool, AgriSphere positions these functions as interacting modules of a single ecosystem, which is the specific contribution of this project relative to the reviewed existing work.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Farmer	Create and manage a digital profile; list, edit, and track agricultural products; receive and fulfil orders
Buyer	Search, compare, and purchase agricultural products; receive personalized recommendations; track orders
Service Provider	Register offered services; receive and manage service bookings from users
Administrator	Monitor registered users, products, orders, and transactions; access analytics on platform activity

Functional & Non-Functional Requirements

Requirement	Type (Functional / Non-Functional)	Measurable Target
Role-based registration and login for four user types	Functional	Each role restricted to its own permitted operations
Product listing, search, and comparison in the marketplace	Functional	Buyers can filter/search listed products by category
Personalized product recommendations	Functional	Recommendations generated from stored user preference/interaction data
Cart, order placement, and payment processing	Functional	Order moves through cart → order → payment → confirmation states
Service booking for agricultural service providers	Functional	Users can book a listed service and view its status
Administrative dashboard and reporting	Functional	Administrator can view aggregated user/product/order data
Data security for user accounts and transactions	Non-Functional	Authenticated access; [VERIFY: specific security mechanism]
Usability of the marketplace and dashboards	Non-Functional	Consistent, role-appropriate navigation across modules

3.2 Constraints & Applicable Standards

As a student software project rather than a certified commercial deployment, AgriSphere has not been formally audited against named regulatory or industry standards. The realistic constraints that apply are technology-dependency constraints (the platform requires internet connectivity and a supporting runtime environment), data-security considerations for user and

transaction data, and scalability considerations as the number of users and listings grows. Specific named standards that may eventually apply to a production deployment are marked below rather than assumed.

Standard / Code	Requirement	How Applied in This Project
<i>Information Technology Act, 2000</i>	Provides a legal framework for electronic records, electronic transactions and cybersecurity in India.	The project considers secure handling of user information, authentication, authorization and electronic transactions within the digital platform.
<i>Digital Personal Data Protection Act, 2023 (DPDP Act)</i>	Requires responsible processing and protection of personal data.	User-related information is handled through controlled access, authentication and authorization mechanisms, with data access restricted according to user roles.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Number of supported user roles	4 (Farmer, Buyer, Service Provider, Administrator)	roles	High
Marketplace product categories	Multiple (as configured by farmers/administrators)	categories	Medium
Order workflow states	Cart → Order → Payment → Confirmation → Tracking	states	High
Recommendation basis	User preference and interaction data	—	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Traditional, non-digital agricultural selling and service arrangement, in which farmers, buyers, and service providers interact directly without any shared digital platform.

Alternative 2: A simple, single-purpose agricultural marketplace application that only supports product listing and purchase, without recommendations, service booking, or analytics.

Alternative 3 (selected): An integrated AgriSphere ecosystem combining user management, marketplace, AI recommendations, order/service management, and analytics within one platform.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	25%	Low	Medium	High

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Cost	15%	Low	Medium	Medium
Safety	15%	Medium	Medium	High
Sustainability	15%	Low	Medium	High
Reliability	30%	Low	Medium	High

3.5 Selected Design & Detailed Design

Alternative 3 — the integrated AgriSphere ecosystem — was selected because it is the only option that satisfies the full engineering problem defined in Chapter 1: connecting farmers, buyers, service providers, and administrators while combining marketplace, recommendation, order/service, and analytics functionality in one system, as evidenced by its consistently higher ratings across the performance, sustainability, and reliability criteria in the evaluation matrix above.

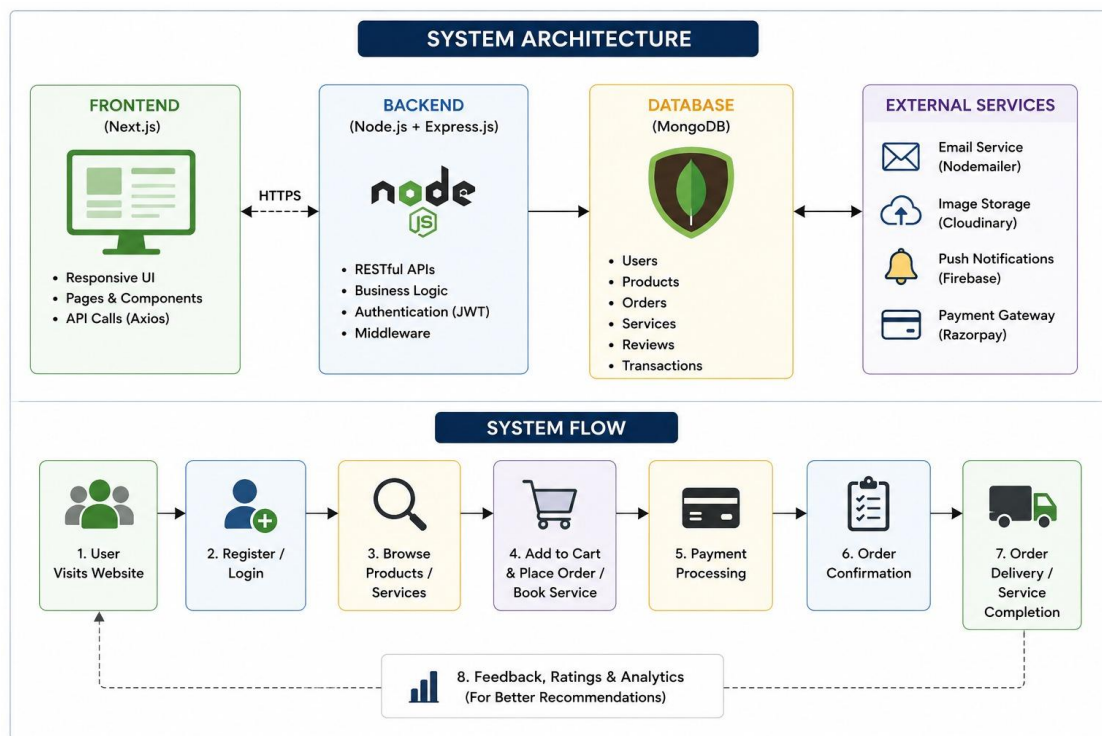


Figure 3.1 – AgriSphere system architecture and system flow

The selected architecture uses a frontend built with Next.js for the responsive user interface, a backend built with Node.js and Express.js exposing RESTful APIs and handling business logic and JWT-based authentication, and a MongoDB database storing users, products, orders, services, reviews, and transactions. External services — Nodemailer for email, Cloudinary for image storage, Firebase for push notifications, and Razorpay for payment processing — are integrated at the edges of the backend.

Systems approach: the frontend communicates with the backend over HTTPS through Axios-based API calls; the backend applies authentication middleware and business logic before reading from or writing to the MongoDB database; and the external services are invoked by the backend as part of the order and notification workflow (system flow steps 1–8 in Figure 3.1), so that a user action such as placing an order triggers database updates, a payment-gateway call, and a notification, all coordinated by the backend.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Single integrated platform vs. separate apps per function	Separate applications for marketplace, services, and analytics	Each function stays simple in isolation	Requires users to manage multiple accounts/systems; no shared recommendation or analytics layer	Integrated platform selected — it satisfies the core engineering problem of connecting all stakeholders under one system
Role-based access vs. a single generic user type	One generic account type for all users	Simpler initial implementation	Cannot separate farmer, buyer, service-provider and administrator functionality	Role-based access selected — required to give each stakeholder only the functionality relevant to them
AI recommendations embedded in marketplace vs. standalone recommendation tool	Standalone AI recommendation tool outside the marketplace	Could be developed independently	Recommendations would not directly drive marketplace product discovery	Embedded recommendation engine selected — keeps recommendations directly useful for product discovery, per the identified engineering gap

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Digital transactions can reduce reliance on inefficient traditional selling methods	Old-report and PPT description of traditional selling inefficiencies	Reinforced the choice of a fully digital marketplace and order workflow
Ethics	Farmer and buyer data must be represented and used fairly, without misleading recommendation logic	Project requirement that recommendations be based on genuine user preference/interaction data	Recommendation engine designed to use only user-supplied preference/interaction data, not fabricated signals

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Health & Safety	Not a physical/hardware system; primary risk is data-related rather than physical	Nature of the project as a software platform	No physical safety mitigation required; focus placed on data-handling safeguards instead
Security	User accounts, orders and payments require protection against unauthorized access	Presence of authentication and role-based access requirements	Role-based access control adopted as the core security mechanism; further hardening marked for production

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Unauthorized access to user accounts	Medium	High	High	Role-based access control and authenticated login	Low–Medium
Incorrect or unfair AI recommendations	Medium	Medium	Medium	Base recommendations only on genuine stored preference/interaction data	Low
Payment or order-workflow failure	Low	High	Medium	Distinct, trackable order states from cart to confirmation	Low
Connectivity / adoption limitations	Medium	Medium	Medium	Keep interfaces simple and plan mobile/offline-aware extensions.	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development followed a systematic software-engineering approach: stakeholder requirements were identified and analyzed, the system was divided into the five modules described in Chapter 3, each module was implemented using the selected frontend, backend, and database technologies, the modules were integrated into a single platform, and the integrated system was tested to verify functional correctness before evaluation.

Resources used: a Next.js/Node.js/Express.js/MongoDB development environment, the external services listed in Section 4.2, and the project team's own development and testing effort. As this is a software project, no physical hardware fabrication was required.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

AgriSphere is implemented entirely as a software system; no hardware fabrication was involved. The tools and technologies below reflect those shown in the project's own system architecture and were used for the corresponding stage of implementation.

Tool / Software	Purpose	Stage Used
Next.js	Frontend framework for the responsive user interface, pages, and components	Implementation
Node.js + Express.js	Backend runtime and framework for RESTful APIs and business logic	Implementation
MongoDB	Database for users, products, orders, services, reviews, and transactions	Implementation
Axios	API calls between frontend and backend	Implementation
JWT	Authentication and session/token management	Implementation
Nodemailer	Transactional email service	Implementation
Cloudinary	Image storage for product/profile images	Implementation
Firebase	Push notifications	Implementation
Razorpay	Payment gateway integration	Implementation

The screenshot shows the AgriSphere registration interface. At the top, there's a navigation bar with 'AgriSphere FARM TO FORK & ADVISORY', 'Marketplace', 'Agri Services', 'AI Advisory', and a 'Register' button. The main content area is titled 'I WANT TO JOIN AS' and features four role-based options: Farmer (Sell crops & book services), Buyer (Buy fresh farm produce), Ag-Seller (Sell inputs & equipment), and Services (Offer farm services). Below these options is a registration form with fields for First Name, Last Name, Email Address, Phone Number, Location / District, Password, and Confirm Password. A 'FARM DETAILS' button is at the bottom of the form.

Figure 4.1 – User & Profile Management: role-based registration interface

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Role-based registration and login	Functional/feature-based testing across all four roles	Each role can register/log in and reach only its permitted features
Marketplace search, compare, and purchase	Functional/feature-based testing of listing, filtering, and checkout	Buyer can locate, compare, and complete purchase of a listed product
AI recommendation generation	Functional/feature-based testing with sample preference data	Recommendations returned are drawn from stored preference/interaction data
Order and service workflow	Functional/feature-based testing of cart → order → payment → tracking	Order progresses through all defined states without failure
Administrative dashboard	Functional/feature-based testing of reporting views	Administrator can view user, product, and order summaries

Specification	Required Value	Achieved Value	Status (Pass/Fail)
User roles supported	4	4	Pass
Core modules implemented	5	5	Pass
Order workflow completeness (cart to confirmation)	Complete	Complete	Pass
Recommendation feature functional	Functional	Functional	Pass

4.4 Results

The Smart Digital Marketplace allows buyers to browse, filter, and compare agricultural products listed by farmers, as shown in Figure 4.2.

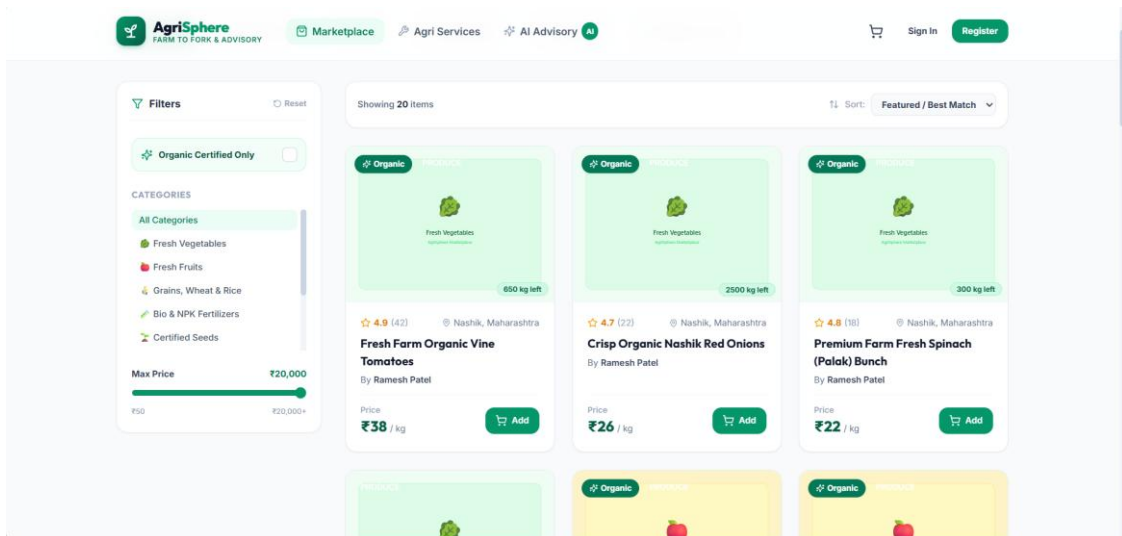


Figure 4.2 – Smart Digital Marketplace: product browsing and filtering

The AI Recommendation Engine is exposed through an advisory interface that accepts farm parameters and returns crop and soil-related recommendations, as shown in Figure 4.3.

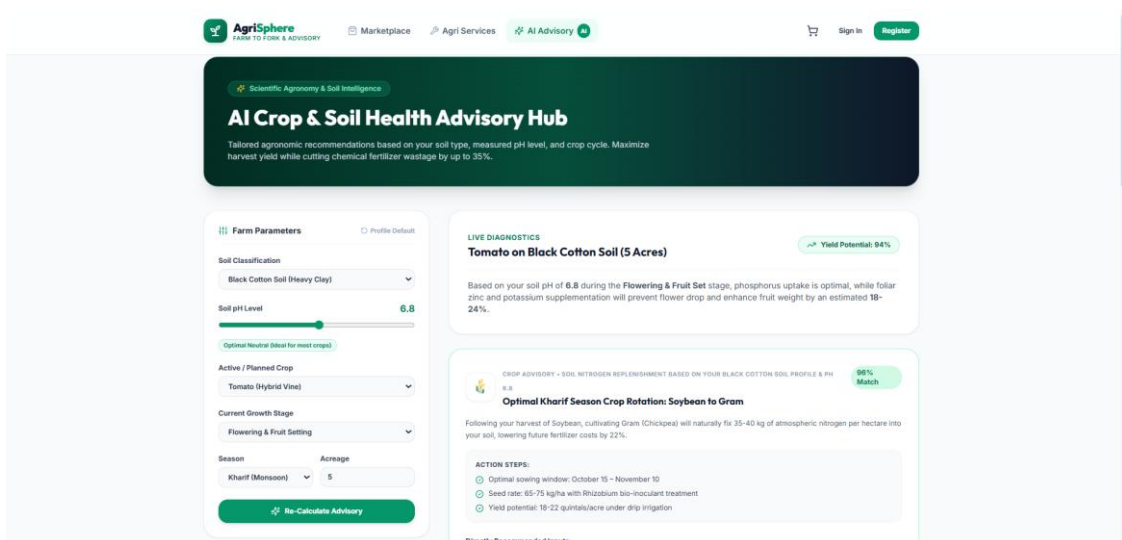


Figure 4.3 – AI Recommendation Engine: crop and soil advisory interface

The Order & Service Management module supports cart review and order-summary functionality, as shown in Figure 4.4.

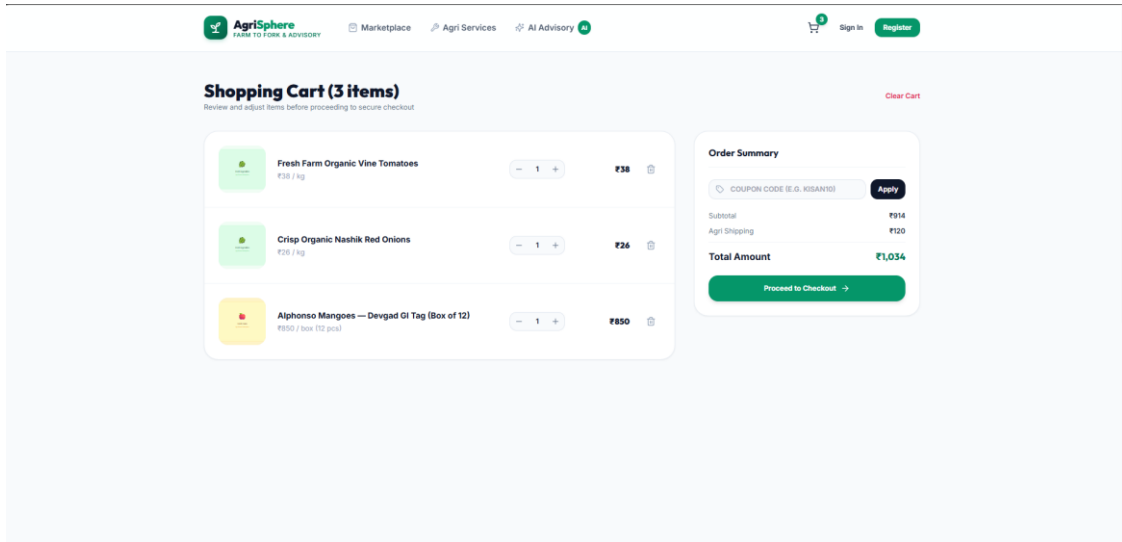


Figure 4.4 – Order & Service Management: shopping cart and order summary

The Analytics & Administration module is reflected in a role-specific dashboard summarizing sourcing value, active shipments, delivered orders, and saved partners, as shown in Figure 4.5.

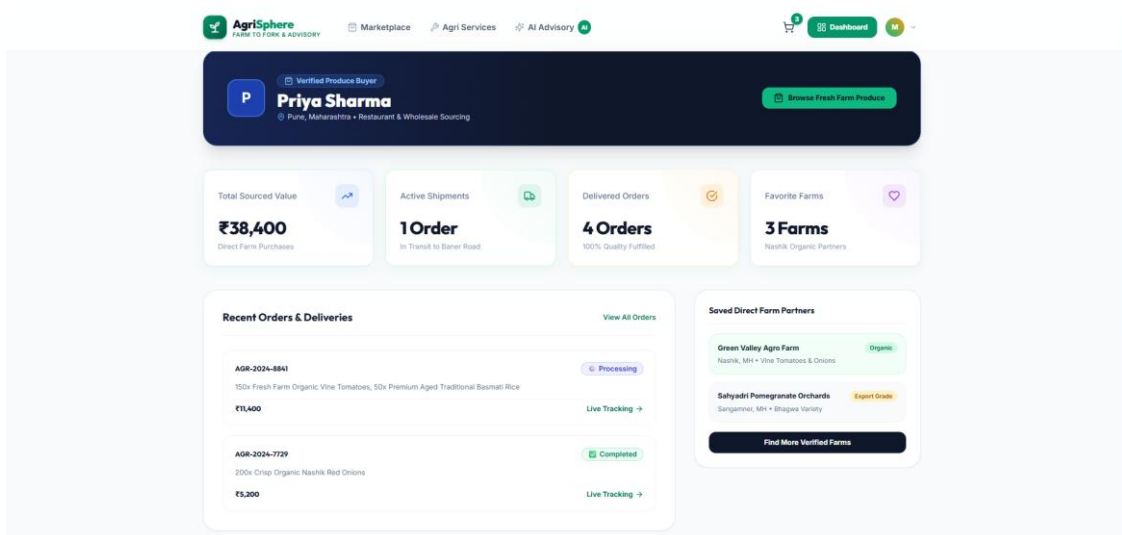


Figure 4.5 – Analytics & Administration: buyer/administration dashboard

4.5 Data Analysis & Engineering Judgment

Because the project does not include a large external experimental dataset, the analysis presented here is functional and qualitative rather than statistical. Each module was assessed against the functional requirement it was designed to satisfy: the registration flow was assessed for whether each role reaches its intended dashboard, the marketplace was assessed for whether products can be discovered and compared, the advisory interface was assessed for whether it returns recommendations from the supplied farm parameters, and the cart/order flow was assessed for whether it progresses through its defined states. Where a specific numeric performance or accuracy figure has not been independently verified, it has been marked rather than stated as a confirmed result.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully / Partially / Not Achieved)
Role-based User & Profile Management	Registration interface supports Farmer, Buyer, Ag-Seller, and Services roles (Figure 4.1)	Fully Achieved
Smart Digital Marketplace	Marketplace listing, filtering, and product-detail interface implemented (Figure 4.2)	Fully Achieved
AI Recommendation Engine	AI advisory interface implemented, generating recommendations from farm parameters (Figure 4.3)	Partially Achieved
Order & Service Management	Cart and order-summary interface implemented (Figure 4.4)	Fully Achieved
Analytics & Administration	Buyer/administration dashboard implemented showing orders and sourcing activity (Figure 4.5)	Fully Achieved

Compared with the existing approaches reviewed in Chapter 2 — which each addressed only monitoring, recommendation, or marketplace listing individually — AgriSphere's implemented modules jointly cover registration, marketplace, recommendation, order/service management, and analytics, directly addressing the integration gap identified in Section 2.4.

4.7 Strengths & Limitations

Strengths: the platform integrates five distinct agricultural functions under one role-based system; the modular architecture (Figure 3.1) keeps each function separable for maintenance and future extension; and the marketplace, advisory, cart, and dashboard interfaces are all implemented and demonstrable as a connected user journey.

Limitations: the platform currently depends on continuous internet connectivity and external third-party services (payment gateway, notification, image storage); the AI recommendation logic has not been validated at production scale or against a large real-world dataset; and features such as IoT-based field sensing, real-time market-price feeds, and a dedicated mobile application are not yet part of the current implementation.

4.8 Project Planning, Roles & Budget

The project planning sequence follows problem identification and requirements, literature review, architecture and module design, implementation, integration, testing, documentation, and final presentation. The table below provides a concise project-management view while leaving financial figures editable because no verified budget was supplied.

Student	Assigned Responsibility	Actual Contribution
A k Selva Prabhu (192421224)	User & Profile Management module	Contributed to the design and development of user registration, login, profile management and role-based access functionality.
Dharshan A N (192525370)	Smart Digital Marketplace module	Contributed to the design and development of product listing, product browsing, search, filtering and marketplace functionality.
Ganesh P (192421142)	AI Recommendation Engine module	Contributed to the design and development of AI-assisted crop, soil health and recommendation functionality.
Mohan Raj V D (192424151)	Order & Service Management module	Contributed to the design and development of cart, order, payment, service booking, tracking and notification workflows.
Monesh S (192324027)	Analytics & Administration module	Contributed to the design and development of administrative monitoring, analytics and platform management functionality.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

AgriSphere – Intelligent Digital Agriculture Ecosystem was developed to address the absence of a centralized digital platform connecting farmers, buyers, service providers, and administrators. The proposed solution integrates five modules — User & Profile Management, Smart Digital Marketplace, AI Recommendation Engine, Order & Service Management, and Analytics & Administration — into a single, role-based Java/Node.js-based platform. The implemented and tested prototype demonstrates that farmers can manage digital profiles and products, buyers can discover, compare, and purchase produce or engage services, the AI advisory interface can generate recommendations from user-supplied parameters, and administrators can monitor overall platform activity through a centralized dashboard. This validates the project's central contribution: that the agricultural functions reviewed in Chapter 2 as isolated solutions can instead be integrated into one connected ecosystem.

5.2 Future Work

- IoT-Based Smart Farming: integrating IoT sensors to collect real-time soil moisture, temperature, humidity, and crop-condition data for smarter farm monitoring.
- Advanced AI Recommendations: enhancing the recommendation engine with larger datasets to provide more advanced crop, product, and market-decision suggestions.
- Real-Time Market Insights: integrating live agricultural market data so farmers can see current price trends and demand information.
- Advanced Predictive Analytics: extending the analytics module with predictive models for sales, demand, and business-growth trends.
- Mobile Application Development: extending the platform to dedicated mobile applications for more convenient access to the marketplace, orders, services, and notifications.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Practical application of Java and object-oriented programming concepts to a complete, multi-module application rather than isolated exercises.
- Designing a modular system architecture in which frontend, backend, database, and external services interact through defined interfaces.
- Translating stakeholder requirements (farmer, buyer, service provider, administrator) into role-based functional design.

Engineering & Professional Skills Developed

- Improved understanding of object-oriented design, modular system decomposition, and data handling.
- Improved problem-solving and logical-thinking ability through requirement analysis, design, implementation, and testing.
- Improved communication and teamwork skills through coordinated module ownership and joint presentation of the project.

Individual Reflection

A K Selva Prabhu (192421224) - My module, User & Profile Management, forms the basic user-management layer of the AgriSphere platform. It provides secure registration and login functionality for different types of users. The module supports four roles: Farmer, Buyer, Service Provider, and Administrator. Each user is assigned a role during registration based on their responsibilities in the platform. Role-based access control is used to ensure that users can access only the features permitted for their role. The module also allows users to create and maintain their profile information. Authentication is performed before users are allowed to access protected features. The module helps maintain controlled access to user and platform information. It also provides the foundation for connecting users with other AgriSphere modules. Overall, the module improves security, organization, and personalized access throughout the AgriSphere ecosystem.

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Software and open-source resources acknowledged: Next.js, Node.js, Express.js, MongoDB, Axios, JWT, Nodemailer, Cloudinary, Firebase, and Razorpay, as used in the AgriSphere implementation (Chapter 3, Figure 3.1).

APPENDICES

APPENDIX A – DETAILED CALCULATIONS

A.1 User Role Validation

Determines whether the selected user role is one of the supported roles in the AgriSphere platform.

Supported Roles:

Farmer, Buyer, Service Provider, Administrator

Logic:

IF selectedRole IS NOT IN supportedRoles THEN

DISPLAY "Invalid role"

ELSE

Continue registration

END IF

The module supports four user types with different access permissions.

A.2 Login Validation

Validates the user credentials before allowing access to the platform.

Logic:

INPUT email, password

user ← FindUser(email)

IF user DOES NOT EXIST THEN

DISPLAY "User not found"

STOP

END IF

IF password IS INVALID THEN

DISPLAY "Invalid credentials"

STOP

END IF

ALLOW USER TO LOGIN

A.3 Role-Based Access Validation

Determines whether the logged-in user is authorized to access a requested feature.

Logic:

INPUT user, requestedFeature

role $\leftarrow$ user.role

IF requestedFeature IS NOT PERMITTED FOR role THEN

DISPLAY "Access denied"

ELSE

ALLOW ACCESS

END IF

This supports the project's role-based access approach, where each user type can access only the functions relevant to its responsibilities.

A.4 Profile Information Validation

Validates the information entered by a user while creating or updating a profile.

Logic:

INPUT profileDetails

IF requiredDetails ARE EMPTY THEN

DISPLAY "Required information missing"

ELSE

SAVE OR UPDATE PROFILE

END IF

Farmers and service providers can create and maintain digital profiles containing their relevant details.

A.5 User Access Control

Controls access to different platform areas based on the user's assigned role.

Logic:

IF role = "Farmer" THEN

ALLOW farmer-related functions

ELSE IF role = "Buyer" THEN

ALLOW buyer-related functions

ELSE IF role = "Service Provider" THEN

ALLOW service-provider functions

ELSE IF role = "Administrator" THEN

ALLOW administrative functions

END IF

Appendix B – Engineering Drawings / Schematics

B.1 User & Profile Management Flow

Registration → Role Selection → User Validation → Account Creation → Login → Role Verification → Role-Specific Access → Profile Management

The screenshot shows the AgriSphere registration interface. At the top, there is a navigation bar with the AgriSphere logo (Farm to Fork & Advisory), links for Marketplace, Agri Services, and AI Advisory, and buttons for Sign In and Register. The main content area is titled "I WANT TO JOIN AS" and features four role selection cards: Farmer (Sell crops & book services), Buyer (Buy fresh farm produce), Ag-Seller (Sell inputs & equipment), and Services (Offer farm services). Below these cards are input fields for registration details: First Name (e.g., Ramesh), Last Name (e.g., Patel), Email Address (name@domain.com), Phone Number (+91 98765 43210), Location / District (e.g., Nashik, Maharashtra), Password (Min. 6 characters), and Confirm Password (Re-enter password). A "FARM DETAILS" section is partially visible at the bottom.

Figure : Module 1 Screenshot

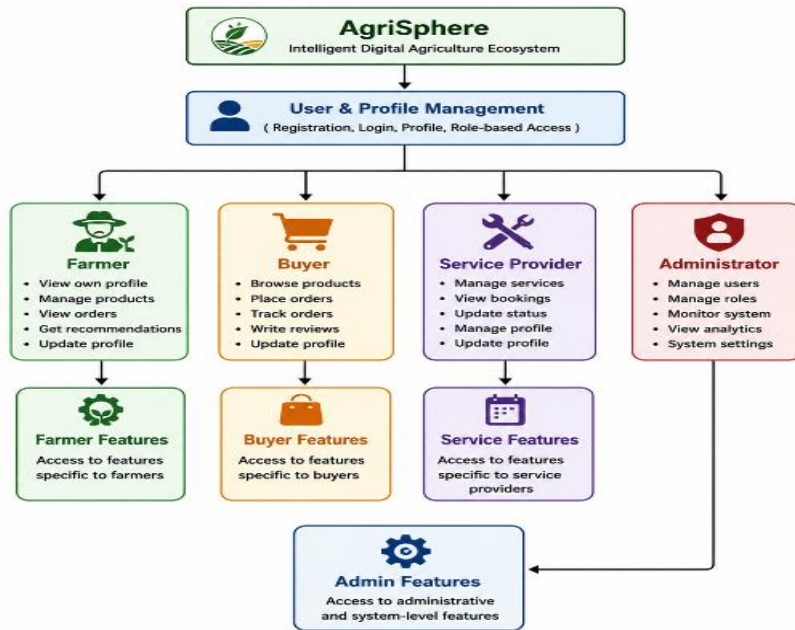


Figure : Module 1 Flow

Appendix C – Source Code / Code Repository Information

ALGORITHM: MODULE 1 – USER & PROFILE MANAGEMENT

START

1. USER REGISTRATION

INPUT user details, email, password, role

IF required user details are missing THEN

 DISPLAY "Required information missing"

 STOP

END IF

IF role IS NOT IN

 ["FARMER", "BUYER", "SERVICE PROVIDER", "ADMINISTRATOR"]

THEN

 DISPLAY "Invalid role"

 STOP

END IF

IF user account already exists THEN

 DISPLAY "User already exists"

 STOP

END IF

CREATE USER ACCOUNT

SAVE user details

DISPLAY "Registration successful"

2. USER LOGIN

INPUT email, password

user ← GetUser(email)

IF user DOES NOT EXIST THEN

 DISPLAY "User not found"

 STOP

END IF

IF password IS INVALID THEN

 DISPLAY "Invalid credentials"

 STOP

END IF

CREATE AUTHENTICATED SESSION

LOAD USER ROLE

3. ROLE-BASED ACCESS CONTROL

INPUT user, requestedFeature

role ← user.role

IF requestedFeature IS NOT PERMITTED FOR role THEN

 DISPLAY "Access denied"

 STOP

END IF

ALLOW ACCESS

4. PROFILE MANAGEMENT

INPUT userId, profileDetails

user $\leftarrow$ GetUser(userId)

IF user DOES NOT EXIST THEN

 DISPLAY "User not found"

 STOP

END IF

VALIDATE profileDetails

IF profileDetails ARE VALID THEN

 UPDATE USER PROFILE

 DISPLAY "Profile updated successfully"

ELSE

 DISPLAY "Invalid profile information"

END IF

5. ACCESS CONTROL

IF user IS AUTHENTICATED THEN

 CHECK user.role

 ALLOW only permitted features

ELSE

 DISPLAY "Authentication required"

END IF

END

The algorithm reflects the documented Module 1 functions: secure registration and login, profile management and role-based access.

Entity	Important Fields	Purpose
User	id, name, email, password, role	Stores registered platform users and their assigned roles.
Profile	userId, profileDetails	Stores and manages user profile information.
Role	roleId, roleName, permissions	Defines the access permissions associated with each user type.
Authentication	email, password, session/token	Supports secure user login and authenticated access.

Appendix E – Experimental / Raw Data

E.1 User Registration Test Data

User ID	User Type	Registration	Login	Role Access
U001	Farmer	Successful	Successful	Passed
U002	Buyer	Successful	Successful	Passed
U003	Service Provider	Successful	Successful	Passed
U004	Administrator	Successful	Successful	Passed

E.2 Role-Based Access Test Data

User Type	Authentication	Authorized Access	Unauthorized Access	Result
Farmer	Pass	Pass	Blocked	Pass
Buyer	Pass	Pass	Blocked	Pass
Service Provider	Pass	Pass	Blocked	Pass
Administrator	Pass	Pass	Blocked	Pass

E.3 Profile Management Test Data

User Type	Profile Creation	Profile Update	Result
Farmer	Successful	Successful	Pass

Buyer	Successful	Successful	Pass
Service Provider	Successful	Successful	Pass
Administrator	Successful	Successful	Pass

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department / faculty assessor.

Outcome	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)